

ENTERPRISE DATA GOVERNANCE FRAMEWORK DESIGN & IMPLEMENTATION FOR E-COMMERCE PLATFORM TRANSFORMATION

DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

2. OBJECTIVES

- Establish clear accountability for data ownership and stewardship
- Improve data quality and trust across all business domains
- Enable regulatory compliance and auditability
- Standardize data governance processes across the enterprise
- Support scalable governance aligned with modern data platforms

3. OPERATING MODEL OVERVIEW

3.1. Governance Model Type

✓ Federated Governance Model

- Central governance defines policies & standards
- Domain teams execute governance within business units

4. GOVERNANCE STRUCTURE

4.1. Organizational Layers

Executive Layer

- Chief Data Officer (CDO)
- Data Governance Council

Strategic Layer

- Data Governance Lead
- Domain Data Owners

Operational Layer

- Data Stewards
- Data Custodians (IT)

Supporting Functions

- Risk & Compliance
- Legal
- Security

4.2. Roles & Responsibilities

Role	Responsibility
Data Owner	Accountable for data domain quality & compliance
Data Steward	Ensures data quality, metadata, definitions
Data Custodian	Technical management of data
Governance Council	Decision-making & policy approval

5. RACI FRAMEWORK

Activity	Owner	Steward	Custodian	Governance Council
Data Definition	A	R	C	I
Data Quality Monitoring	A	R	C	I
Access Control	A	C	R	I
Policy Approval	R	C	I	A
Metadata Management	A	R	C	I
Issue Resolution	A	R	C	I

(R = Responsible, A = Accountable, C = Consulted, I = Informed)

6. CORE GOVERNANCE PROCESSES

6.1. Data Quality Management

- Define critical data elements (CDE)

Determination criteria:

- Regulatory/reporting impact
- Revenue/payment impact
- Customer experience impact
- Fraud/risk impact
- Operational dependency across domains

Example of e-commerce CDE: customer_id, email, order_id, payment_status, sku, vendor_id.

- Establish data quality rules & thresholds
- Monitor via dashboards
- Issue management workflow:
 - Detect → Log → Assign → Resolve → Validate → Close

6.2. Metadata Management

- Maintain enterprise data catalog
- Define business glossary
- Track data lineage (source → transformation → consumption)

6.3. Data Access & Security

- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC) (optional advanced)
- Access request & approval workflow
- Periodic access review

6.4. Data Lifecycle Management

- Define lifecycle stages:
 - Create → Store → Use → Archive → Delete
- Retention policies based on regulatory requirements

6.5. Data Classification

- Define classification levels:
 - Public
 - Internal
 - Confidential
 - Restricted (PII / highly sensitive)

7. GOVERNANCE ARTEFACTS

Mandatory Artefacts:

- Data Governance Policy
- Data Standards
- Data Quality Rules
- Data Classification Policy
- Data Access Policy
- Data Dictionary / Glossary
- Data Lineage Documentation
- Governance Control Register

8. TECHNOLOGY & TOOLING

8.1. Data Governance Tools

- Data Catalog: Collibra / Alation / Atlan / Microsoft Purview
- Data Quality: Informatica / Great Expectations

8.2. Data Platforms

- Data Warehouse (Snowflake, BigQuery)
- Data Lake / Lakehouse (Databricks)

8.3. Integration

- Metadata ingestion from ETL pipelines
- Automated lineage tracking

9. INTEGRATION WITH DATA ARCHITECTURE

- Governance embedded in:
 - Data ingestion pipelines
 - Transformation layers
 - BI / reporting layer
- Controls applied at:
 - Data creation
 - Data storage
 - Data access

10. COMPLIANCE & RISK MANAGEMENT

- Align with regulatory frameworks (e.g., GDPR principles)
- Implement audit trails
- Periodic compliance assessment
- Risk identification and mitigation

11. KPIs & SUCCESS METRICS

Metric	Description
Data Quality Score	>95% of data meeting quality standards
Issue Resolution SLA	>95% of issues are resolved within the SLA according to severity
Data Availability	>99.5% availability for Tier-1 datasets during business hours / agreed SLA
Metadata Completeness	>98% of critical datasets documented in the catalog
Access Review Completion	100% quarterly completion for privileged roles

KPI Definitions:

KPI	Formula (Example)	Target Phase 1–2	Target Steady State
DQ Score	% CDE records that pass all quality rules	≥90%	≥95%
Metadata Completeness	% critical data assets with owner, steward, definition, classification, lineage	≥80%	≥98%
Owner Coverage	% critical data assets with Data Owner	≥95%	100%
Issue Resolution SLA	% of issues are resolved within the SLA according to severity	≥80%	≥95%
Access Review Completion	% quarterly access reviews completed on time	≥90%	100%

12. IMPLEMENTATION ROADMAP**Phase 1: Assessment**

- Current state analysis
- Gap identification

Phase 2: Design

- Define TOM
- Develop policies & standards

Phase 3: Pilot

- Implement in selected domain

Phase 4: Rollout

- Scale across enterprise

Phase 5: Optimization

- Continuous improvement

13. CHANGE MANAGEMENT & ADOPTION

- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

14. ESCALATION FRAMEWORK

- Data issues escalated:
 - Steward → Owner → Governance Council
- SLA for resolution defined

SLA Table (Example):

Severity	Initial Response	Target Resolution
Critical	< 4 jam	< 24 jam
High	< 1 hari kerja	< 5 hari kerja
Medium	< 2 hari kerja	< 10 hari kerja
Low	< 5 hari kerja	< 30 hari kerja

15. REVIEW & CONTINUOUS IMPROVEMENT

- Quarterly governance review
- Annual TOM update
- Continuous KPI monitoring

16. APPENDICES

Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Timeliness

Appendix B: Domain-Centric Governance

Domain	Data Owner	Steward	Custodian
Customer	Head of CRM/Marketing	CRM Steward	Data Platform/IT
Product	Merchandising Lead	Catalog Steward	IT
Order & Transaction	Operations/Payments Lead	Order Steward	IT
Vendor	Supply Chain Lead	Vendor Steward	IT

Appendix C: Sample Workflow

- Data Issue Management Workflow Diagram

Visual Diagram (Architecture + Governance Flow)

